CATHOLIC JUNIOR COLLEGE
JC2 PRELIMINARY EXAMINATIONS
Higher 2

CANDIDATE NAME			
CLASS	2T	**INDEX NUMBER**	

BIOLOGY

9648/02

Paper 2: Core Paper

26th August 2014
2 hours

Additional Materials: Writing Paper

READ THESE INSTRUCTIONS FIRST

Write your index number and name on all the work you hand in.
Write in dark blue or black pen on both sides of the paper. **[PILOT FRIXION ERASABLE PENS ARE NOT ALLOWED]**
You may use a soft pencil for any diagrams, graphs or rough working.
Do not use staples, paper clips, highlighters, glue or correction fluid.

There are two sections in this paper.

Section A]
Answer all questions

Section B]
Answer **ONE** question. Answer each part on a **separate** piece of paper.

At the end of the examination, fasten all work securely together.
The number of marks is given in brackets [ ] at the end of each question or part of the question.

For Examiner's Use	
Section A	**80**
1 [10]	
2 [10]	
3 [10]	
4 [10]	
5 [10]	
6 [10]	
7 [10]	
8 [10]	
Section B	**20**
9a / 10a [7]	
9b / 10b [7]	
9c / 10c [6]	
TOTAL	**100**

Section A

Answer **all** questions in this section.

1 All cells are surrounded by a cell surface membrane. There are also membrane systems within the cell, as well as several types of membrane-bound-organelles. Some of these have a single membrane surrounding them, while others have two.

All cell membranes have a similar basic structure but the proportions of the various molecules present vary between membranes and even between the inner and outer layers of a membrane.

Fig. 1.1 shows a generalised cell surface membrane.

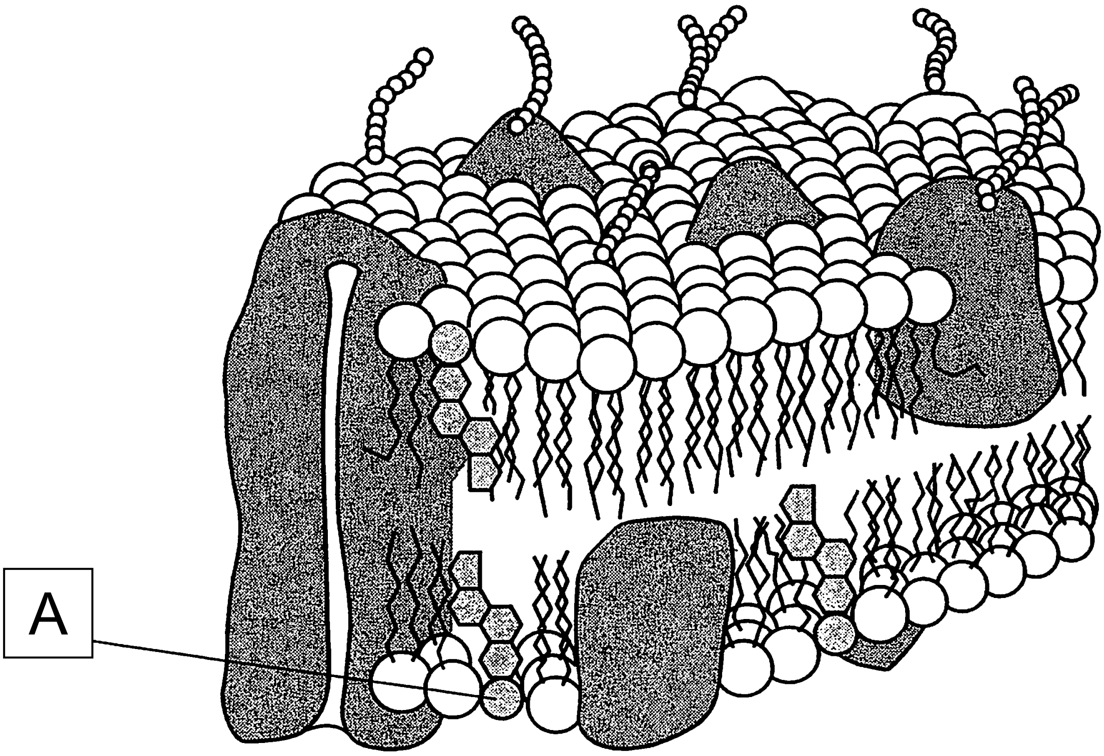

Fig. 1.1

With reference to Fig. 1.1,

(a) (i) identify the molecule A and describe its function. [2]

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(ii) state one way in which the outer surface membrane is different from the inner surface membrane. [1]

(iii) state one functional property resulting from this difference [1]

(b) Table 1.1 shows the approximate percentage of protein in different membranes.

Table 1.1

Organelle	Membrane	Protein as a percentage of dry weight.
chloroplast	outer membrane	54
chloroplast	inner membrane	70
mitochondrion	outer membrane	55
mitochondrion	inner membrane	80

(i) Using data from Table 1.1, describe **one similarity** in the distribution of proteins within the two membranes of chloroplast and mitochondrion and suggest a reason. [3]

Similarity	
Reason	

(ii) Describe and explain one physical way in which the inner membrane of a mitochondrion is adapted to to its function. [3]

[Total: 10]

2 Protein kinases are a class of enzymes that are involved in the triggering of the cell cycle. Kinases add phosphate to molecules, and the modification can serve as a "switch" to turn events in the cell on or off. Cdk or cyclin dependent kinases (Cdk) regulate the cell cycle.

Fig. 2.1 shows a representation of Cdk

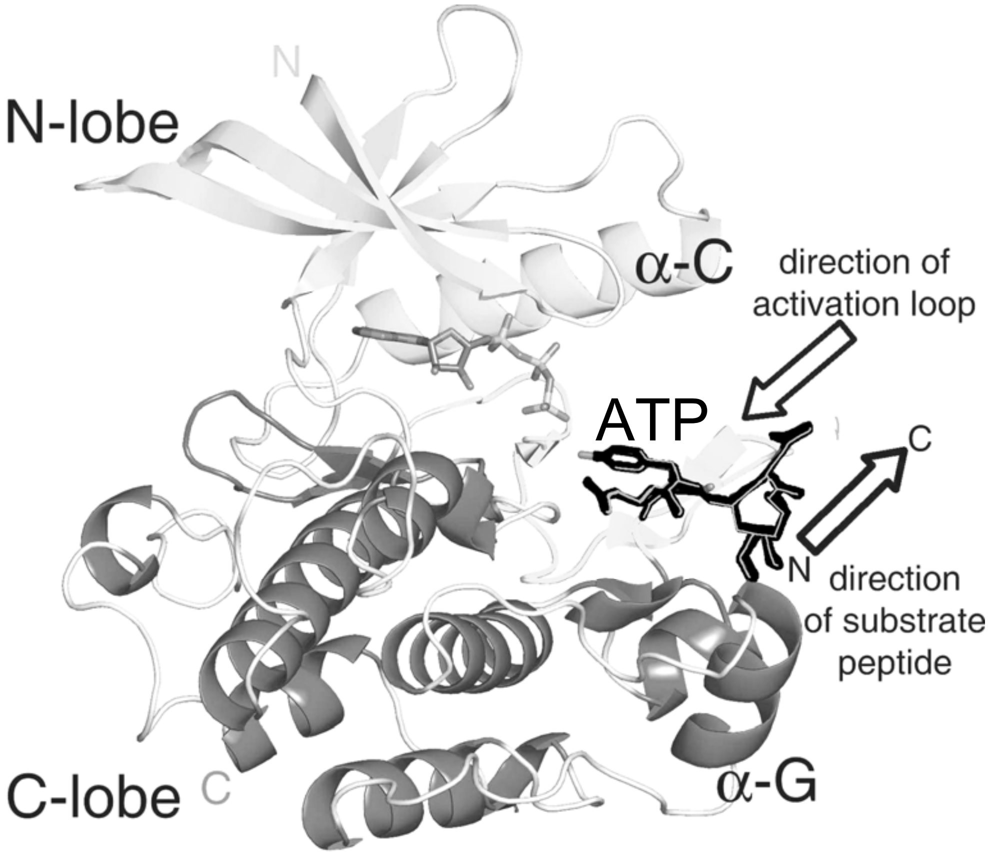

Fig. 2.1 Anatomy of a Cdk catalytic domain. The substrate peptide, docked onto the activation loop, and an ATP analog are shown in black. α-Helix-C, which contains residues critical for catalysis, and α-helix-G, which is involved in docking protein substrates, are labeled.

(a) Describe how specialised enzymes are structurally specific to their substrate [2]

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____

With reference to Fig. 2.1,

(b) explain the specific significance of the following parts of protein kinase [2]

Feature	**Significance**
N-Lobe and C-Lobe	
α-C and α-G	Catalytic R Groups and binding of the R groups
directions of activation loop and substrate peptide	

Cyclin-dependent kinases (Cdk) are a family of protein kinases first discovered for their role in regulating the cell cycle.

(c) Suggest how Cdk may be regulated in its action [1]

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Cdk works always in conjunction with cyclins (another cofactor in the regulation of the cell cycle) the complexes that are formed when these two are combined makes Cdk active.
Fig. 2.2 shows the relative levels of active Cdk (protein kinase activity) during the cell cycle.

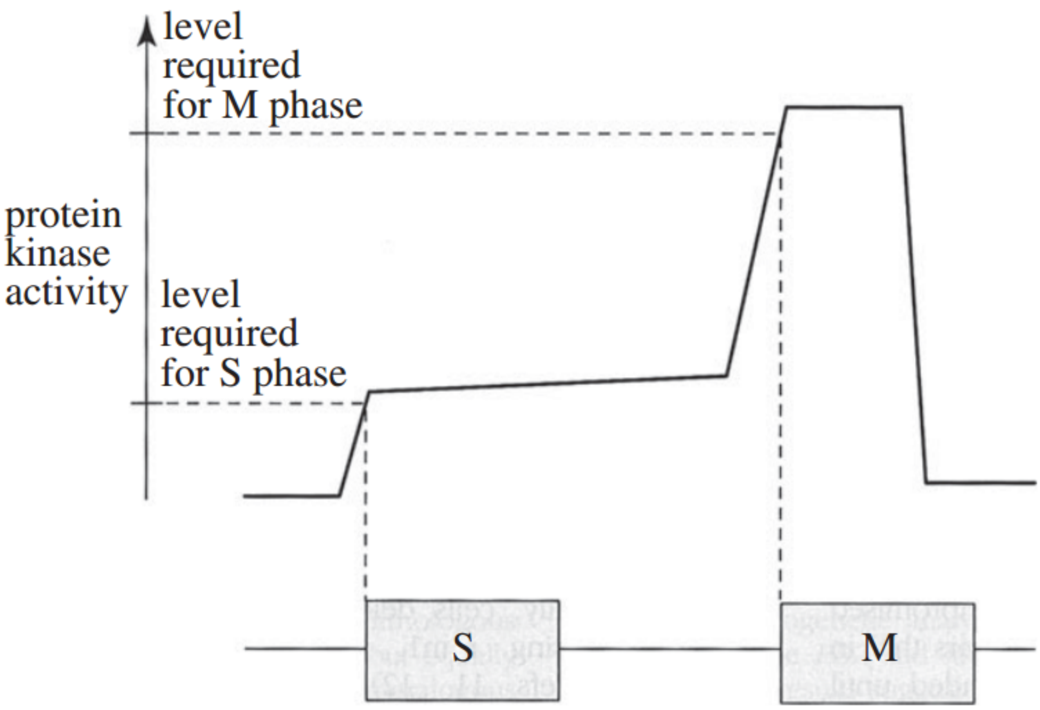

Fig 2.2

(d) Explain what takes place in the gap in-between S phase and M phase (cell division). [3]

(e) Explain how Cdk is used in the regulation of the cell cycle. [2]

[Total: 10]

3 Fig. 3.1 shows the elongation phase of the cellular process of transcription.

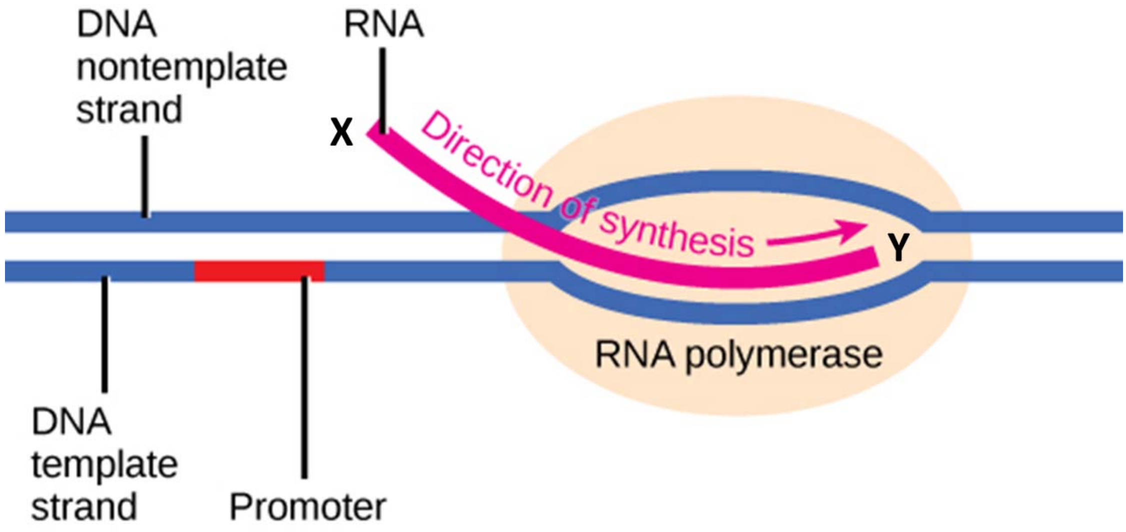

Fig. 3.1

(a)(i) Identify the directionality of the RNA transcript at ends X & Y. [1]

X: ____________________________

Y: ____________________________

(ii) State what are the substrates which are not shown in Fig. 3.1 that RNA polymerase allows into its active site for the process of elongation to take place. [1]

____

(iii) The elongation of RNA transcripts requires energy. Explain how this energy is derived. [1]

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Fig. 3.2 shows that in breast cancer cells, there is an overexpression of human epidermal growth factor receptor 2 (HER2) protein.

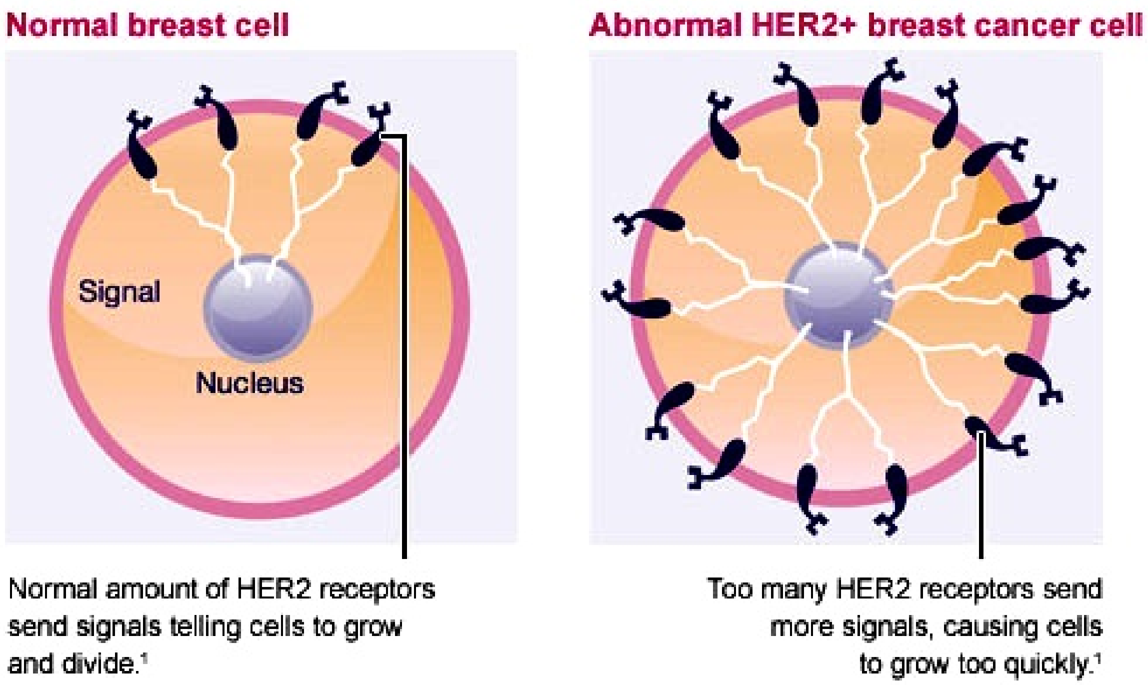

Image reproduced from http://nanaymiriam.wordpress.com/2011/04/09/curcumin-and-first-line-herceptin-monotheraphy/

Fig. 3.2

(b) (i) With reference to Fig. 3.2, suggest why there is an overexpression of HER2 protein. [2]

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(ii) Describe and explain another cause not shown in Fig. 3.2 which could result in an overexpression of the HER2 protein. [2]

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(iii) Suggest with reasons whether HER2 is an oncogene or a tumour suppressor gene. [2]

Fig. 3.3 shows the breast cancer drug, Herceptin® which is a monoclonal antibody that binds to HER2 on the cell surface membrane.

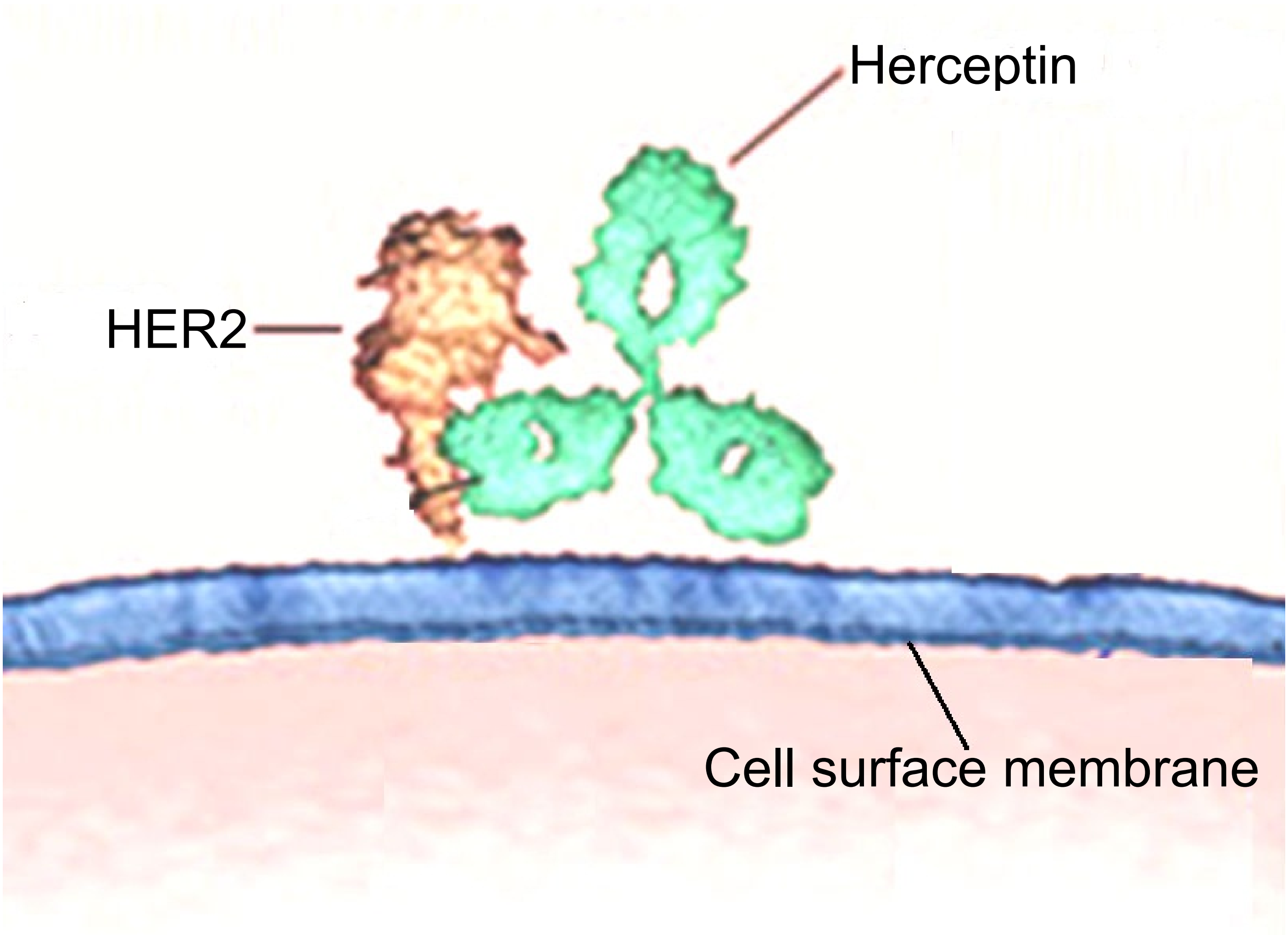

Image modified from http://www.perjeta.com

Fig. 3.3

(c) With reference to Fig. 3.3, suggest how Herceptin® stifles the development of cancer. [1]

[Total: 10]

4 Fig. 4.1 shows regulation of the arabinose operon. The structural genes of the arabinose operon are *araB*, *araA* & *araD*. These structural genes code for enzymes that are responsible for the generation of xylulose 5-phosphate from arabinose. The initiator sites comprise the operator site (*araO*) and the promoter site (*araI*), whereas the activator protein is encoded by *araC*.

In the presence of arabinose, AraC protein and the CAP-cAMP complex bind to different regions of *araI* resulting in the transcription of its structural genes (Fig. 4.1A). In the absence of arabinose, AraC protein binds to both *araI* & araO, which results in a DNA loop that prevents transcription from taking place (Fig. 4.1B).

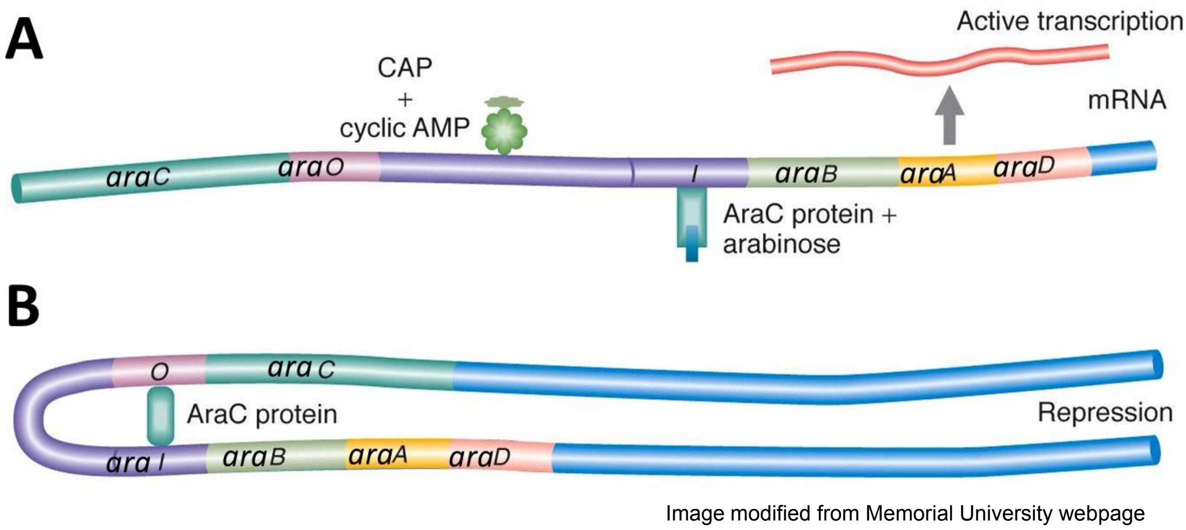

Fig. 4.1

(a) State and explain which type of metabolic reaction arabinose operon is involved in. [2]

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The arabinose operon is controlled by both positive & negative regulation.

With reference to Fig. 4.1, explain how the arabinose operon exhibits:

(b) **(i)** positive regulation [2]

(ii) negative regulation [2]

(c) A mutation resulted in the arabinose operon system to be permanently repressed.

Suggest where and what type of mutation could have occurred. [1]

The Middle East Respiratory Syndrome (MERS) is a viral respiratory disease that is caused by a coronavirus known as MERS-CoV. In 2012, the first case of MERS was reported in Saudi Arabia. Thereafter, other countries near and within the Arabian Peninsula also reported confirmed cases of MERS. Moreover, infected individuals from the affected Middle Eastern countries spread the virus to other countries when they travelled overseas. Individuals, who were infected with MERS-CoV, usually go on to develop severe acute respiratory illness. To date, more than 100 people have died from MERS.

Fig. 4.2 shows the life cycle of MERS-CoV.

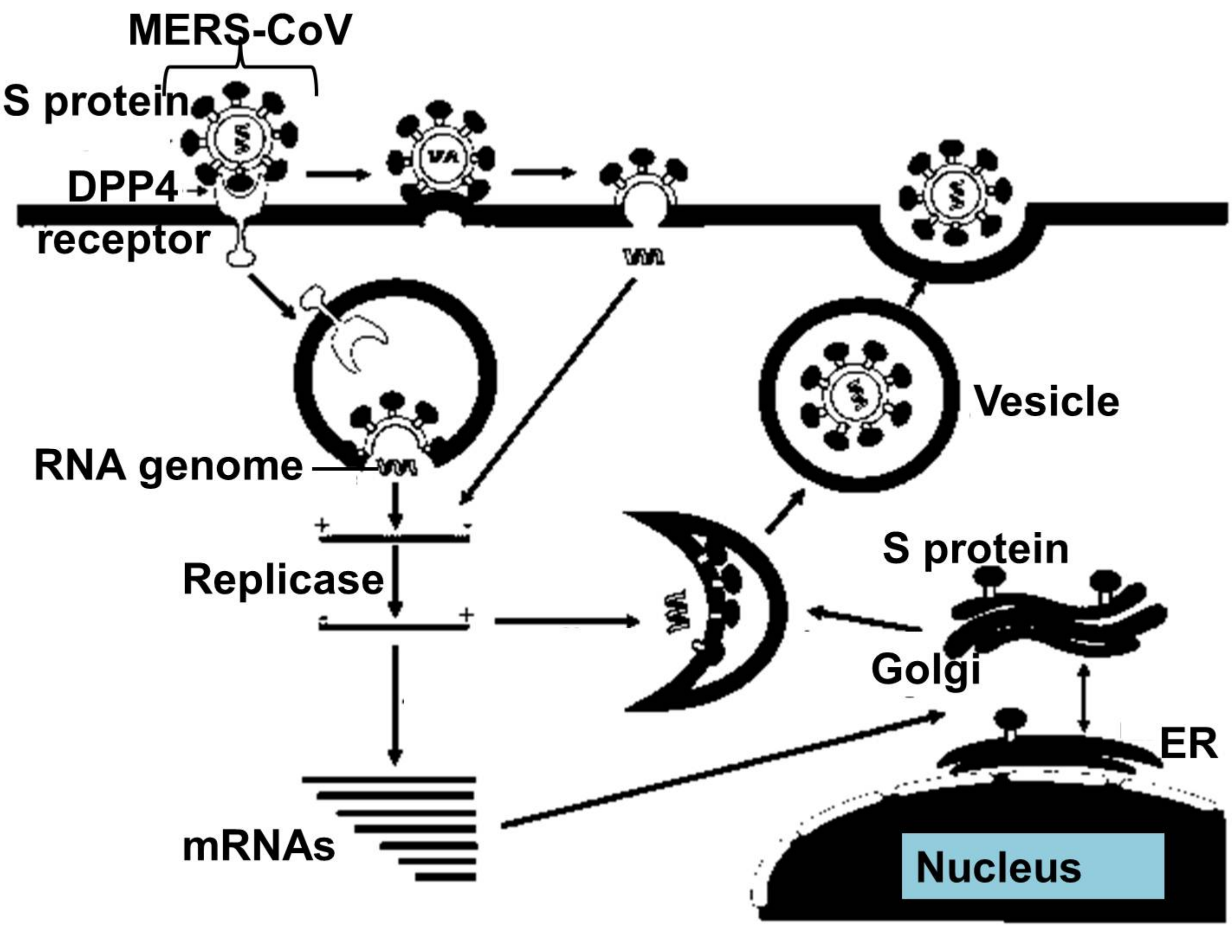

Image reproduced from the article: Middle East respiratory syndrome coronavirus (MERS-CoV): challenges in identifying its source and controlling its spread.

Fig. 4.2

(d) Assuming you are a researcher interested in designing drug to treat MERS, suggest with reference to Fig. 4.2 how you would go about inhibiting the MERS-CoV life cycle. [2]

Camels in Qatar, Oman, Egypt and Saudi Arabia were found to be also infected with MERS-CoV and it has been hypothesised that MERS-CoV spreaded to humans who came into close contact with infected camels.

(e) Suggest how MERS-CoV evolved to infect both humans and camels. [1]

[Total: 10]

5 Fig. 5.1 shows the binding of a general transcription factor to part of the promoter.

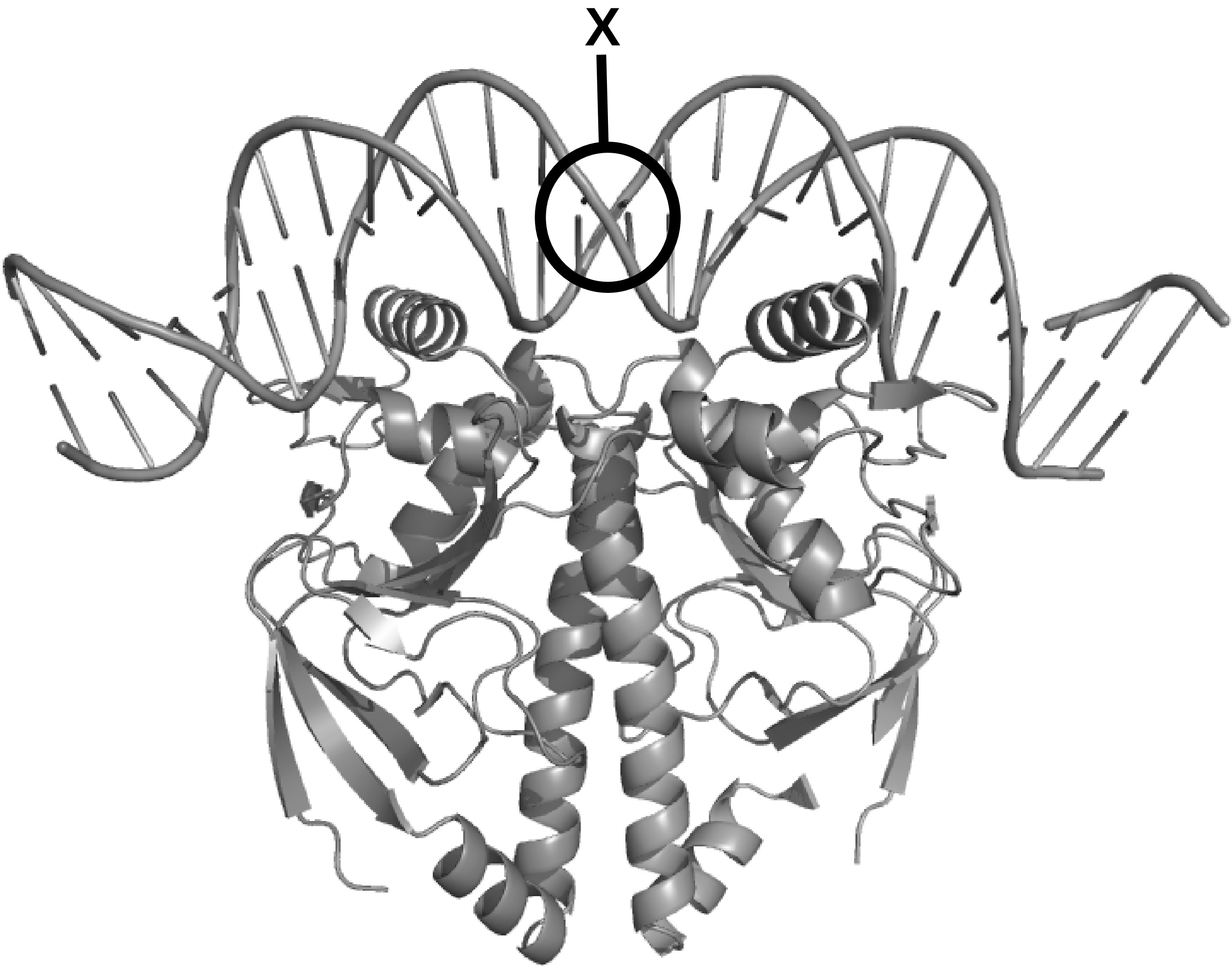

Fig. 5.1

(a) State the function of the general transcription factor. [1]

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(b) Explain how a substitution mutation occurring at X (circled region) would affect gene expression. [2]

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(c) Describe how proteins, other than general transcription factors, can increase the rate of transcription of a gene beyond basal level. [2]

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Fig 5.2 below shows a Philadelphia chromosome which is the result of reciprocal translocation between chromosome 9 and chromosome 22. This chromosomal abnormality is often associated with chronic myeloid leukemia (CML) which is cancer of the white blood cells.

- Abl gene expresses a membrane-associated tyrosine kinase which activates a number of cell cycle control proteins and enzymes.
- It is found that bcr/abl fusion gene expresses a mutant tyrosine kinase that is continuously activated which leads to unregulated cell division.
- Tyrosine kinase activity of bcr/abl fusion protein is elevated relative to wild-type abl protein.
- In addition, cells with bcr/abl fusion protein are resistant to apoptosis induced by DNA damage.

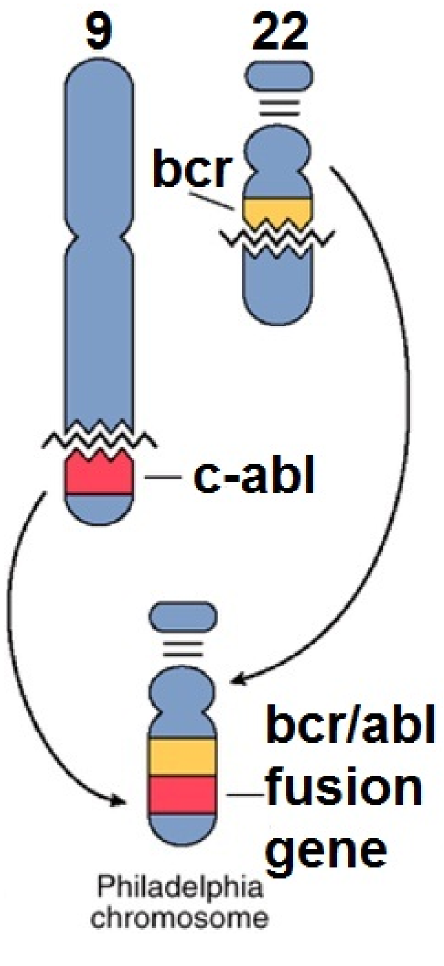

Fig. 5.2

(d) With reference to Fig. 5.2 and the information provided,

(i) state the resultant type of mutation and explain your answer. [2]

(ii) Explain how mutant bcr/abl gene contributes to cancer development in CML. [2]

(e) Suggest a therapeutic strategy that can be used to treat CML caused by Philadelphia chromosome. [1]

[Total: 10]

6 In the inheritance of feather colour in chickens, Individuals carrying the dominant allele, **I**, have white plumage even if they also carry the dominant allele, **C**, for coloured plumage.

White Leghorn chickens have the genotype **IICC**, while white Wyandotte chickens have the genotype **iicc**.
A white Leghorn is crossed with a white Wyandotte and the F_2 generation yield 27 chickens with white plumage and 7 chickens with coloured plumage.

(a) Use a genetic diagram to show the expected genotypes and phenotypic ratios in the F_2 generation. [4]

(b) (i) State the name for this type of interaction between gene loci. [1]

(ii) Explain why gene locus C cannot be expressed in the presence of dominant allele I.[1]

(c) Explain how you could determine that a chicken with coloured plumage was homozygous at both loci. [3]

(d) Given that the calculated chi-square value for this cross is 0.098 and the chi-square table given below. Conclude if the observed results follow the expected phenotypic ratio at 5% level of significance. [1]

Chi-square table

Degrees of Freedom *(df)*	**Probability** (*p*)										
	0.95	0.90	0.80	0.70	0.50	0.30	0.20	0.10	0.05	0.01	0.001
1	0.004	0.02	0.06	0.15	0.46	1.07	1.64	2.71	3.84	6.64	10.83
2	0.10	0.21	0.45	0.71	1.39	2.41	3.22	4.60	5.99	9.21	13.82
3	0.35	0.58	1.01	1.42	2.37	3.66	4.64	6.25	7.82	11.34	16.27
4	0.71	1.06	1.65	2.20	3.36	4.88	5.99	7.78	9.49	13.28	18.47

[Total: 10]

7 Parkinson's disease is a neurodegenerative disorder which leads to progressive deterioration of motor function due to loss of dopamine-producing brain cells.

A substance called dopamine acts as a messenger between two brain areas - the substantia nigra and the corpus striatum - to produce smooth, controlled movements.

Most of the movement-related symptoms of Parkinson's disease are caused by a lack of dopamine due to the loss of dopamine-producing cells in the substantia nigra.

When the amount of dopamine is too low, communication between the substantia nigra and corpus striatum becomes ineffective, and movement becomes impaired; the greater the loss of dopamine, the worse the movement-related symptoms.

Other cells in the brain also degenerate to some degree and may contribute to non-movement related symptoms of Parkinson's disease.

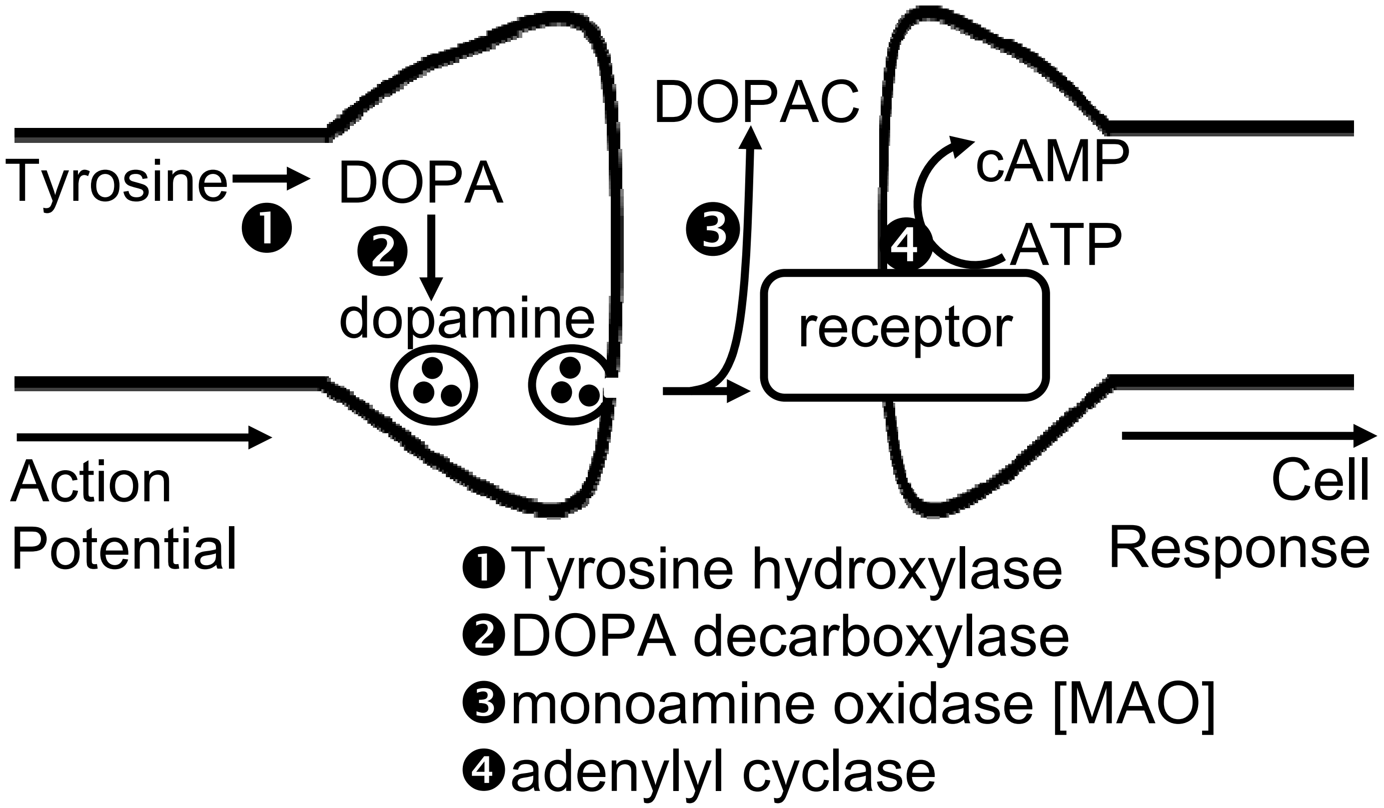

Fig 7.1

(a) Explain the role of dopamine in normal motor functioning [2]

(b) With reference to Fig. 7.1 suggest **three** theoretical treatments of Parkinson's disease with the use of 'designer' enhancing or inhibiting drugs [3]

Fig. 7.2 and 7.3 compare the cells within the substantia nigra in patients with Parkinson's disease and normal individuals.

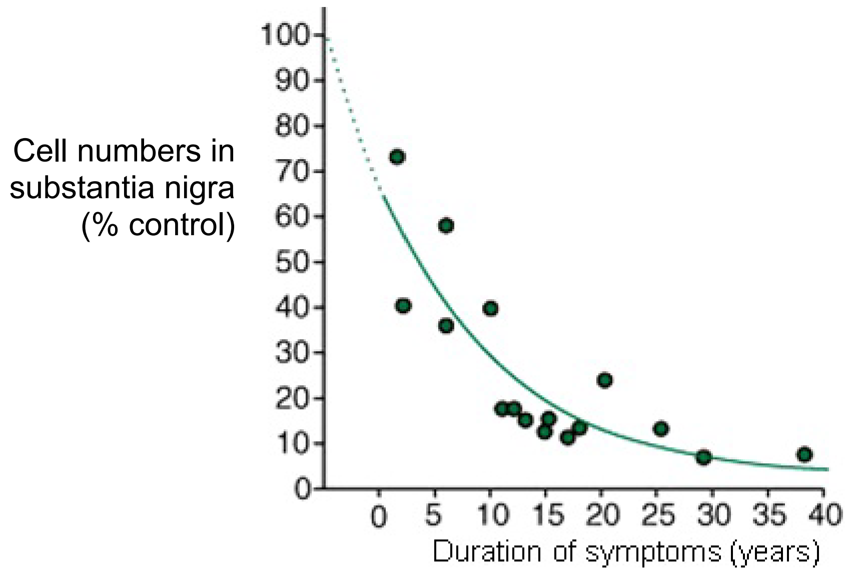

Fig. 7.2 Cell numbers in substantia nigra in patients with Parkinson's disease compared over time

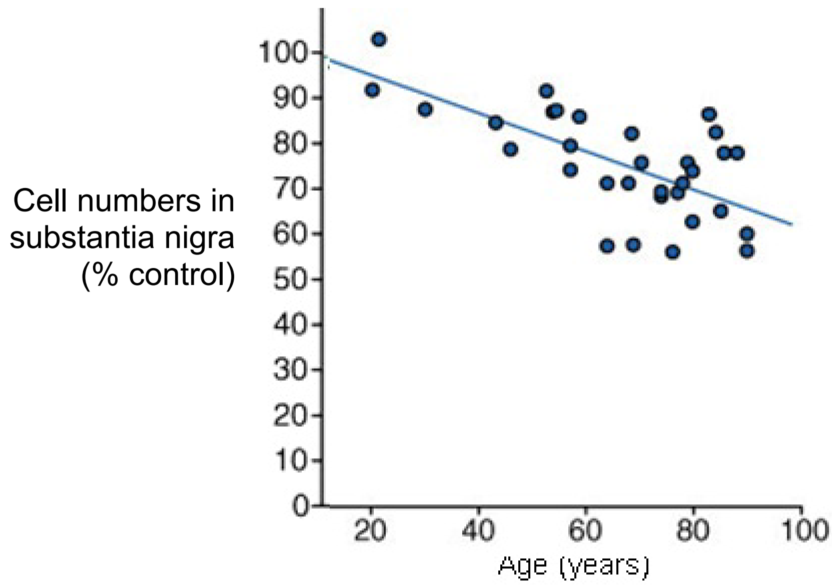

Fig. 7.3 Cell numbers in substantia nigra in normal individuals compared over time

(c) With reference to Fig. 7.1, 7.2 and 7.3 explain the factors leading to Parkinson's disease [3]

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(d) With reference to Fig. 7.2**,**

suggest why upon the onset of Parkinson's disease at year '0' the cell count in substantia nigra is at 70%. [2]

[Total: 10]

8 Cichlids are a fast expanding group of African freshwater fish which is both large and diverse. At least 1,650 species have been scientifically described, making it one of the largest vertebrate families. New species are discovered annually, and many species remain undescribed.

Fig. 8.1

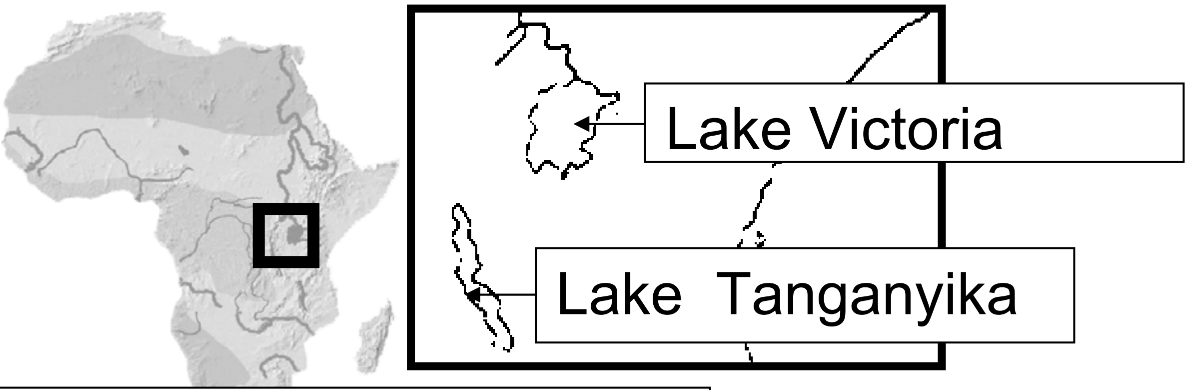

Fig 8.2

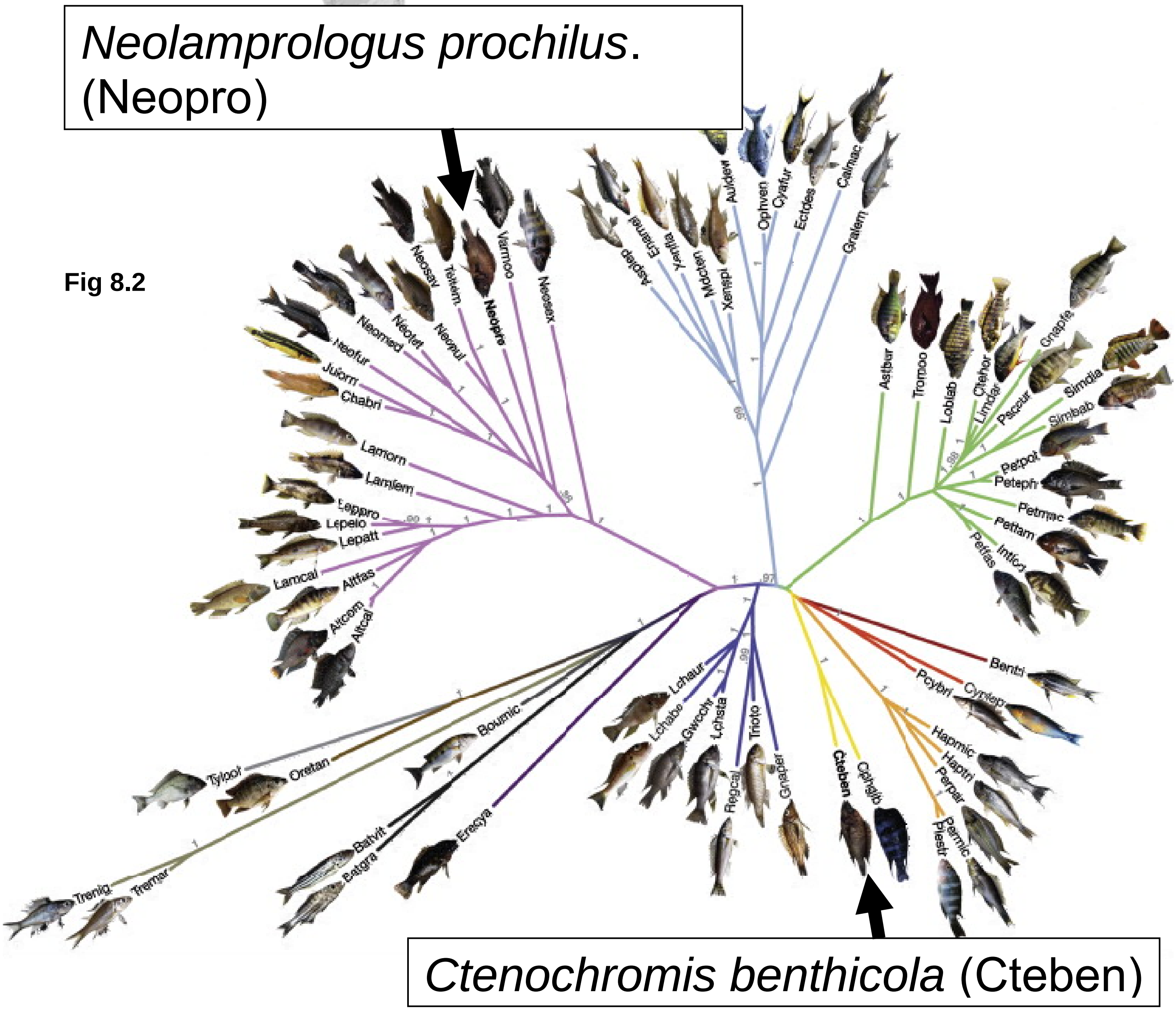

Fig 8.1 Map of East Africa showing the Great Lakes. Lake Victoria (LV) and Lake Tanganyika (LT) which is the oldest lake in East Africa

Fig 8.2 The Cichlid Species DNA comparative phylogeny of the 71 cichlid species (**Family**: *Cichlidae* ; **Order** *Perciformes*) of Lake Tanganyika

(a) With reference to Fig. 8.1 and 8.2, complete the taxonomy of 'Neopro' : [1]

Genus : ____

Species : ____

(b) Explain the possible evolutionary processes that could have led to the many species seen in Fig 8.2. [3]

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(c) Explain how DNA homology is used to predict phylogeny. [2]

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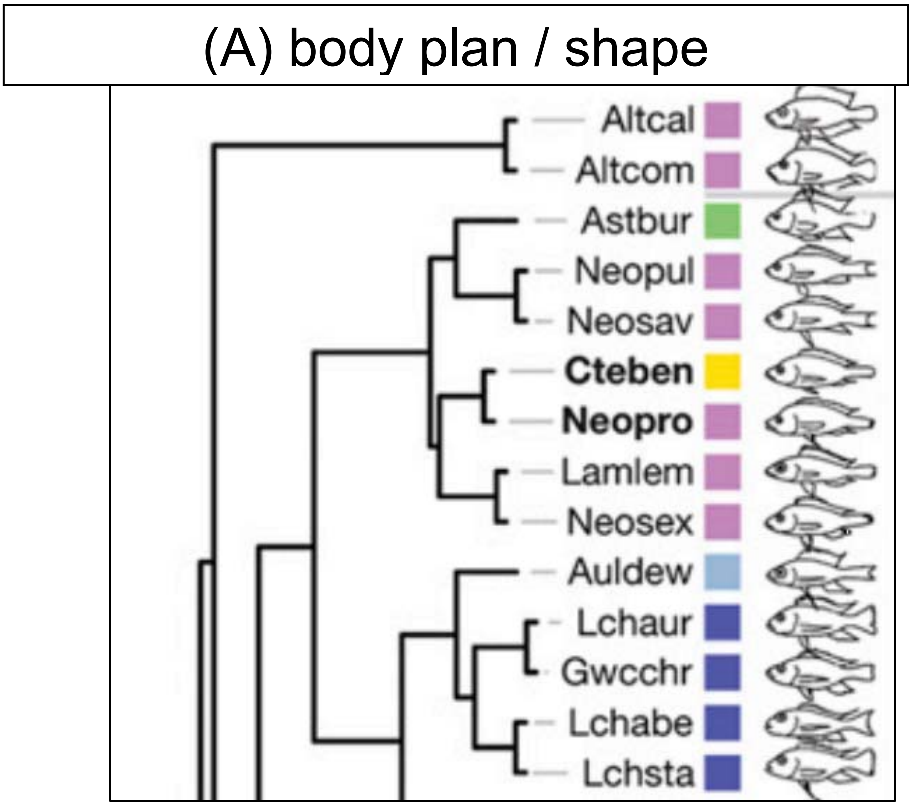
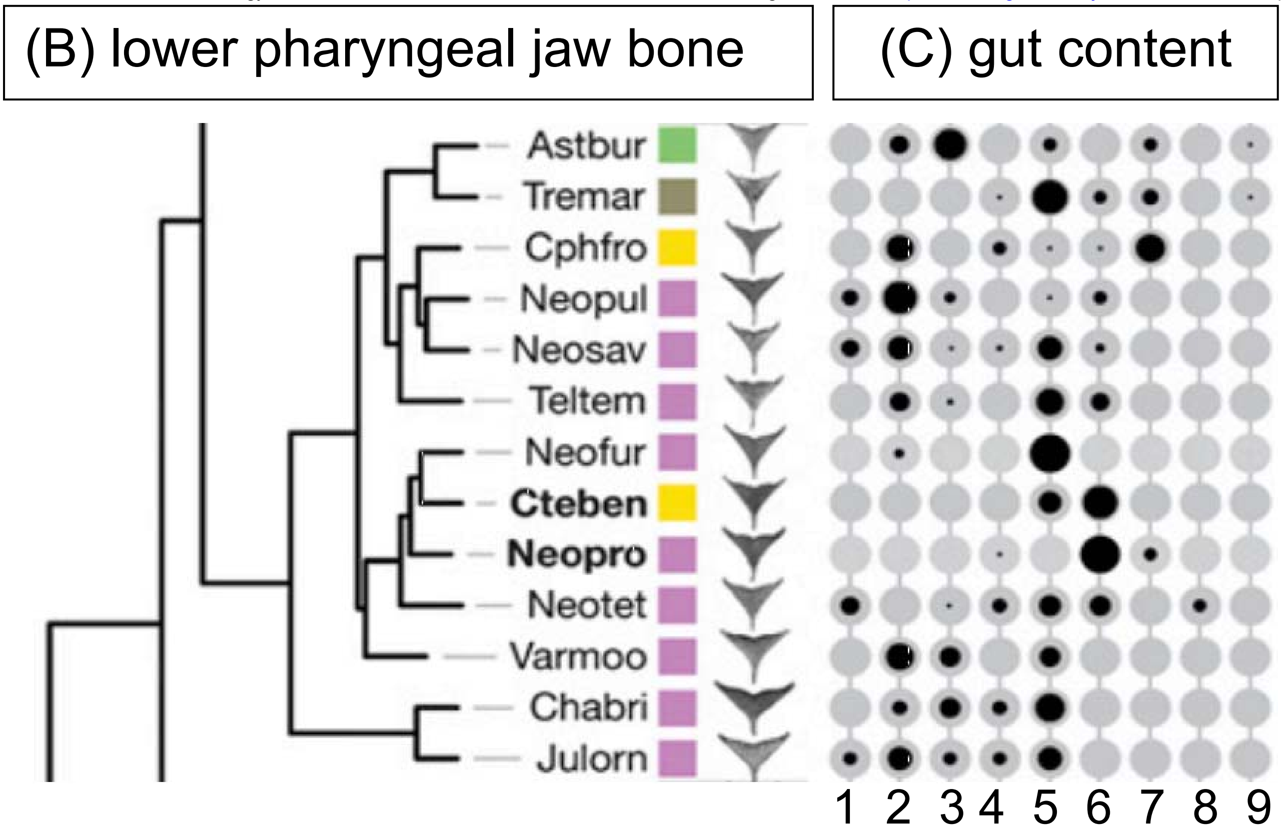
Fig. 8.3
(A) Portion of the Cluster analysis * proposing phylogenetic relationships on the basis of 17 landmarks on body shape. Outlines in (A) are based on real photographs

*Cluster Analysis is a systematic comparison of parameters that allows a conclusion on relationships between specimens.





1- sand
2- aufwuchs- surface overgrowth
3- plants
4- mollusks
5- insects
6- crustaceans
7- fish remains
8- fish eggs
9- fish scales

Fig. 8.4
(B) Portion of the Cluster analysis proposing phylogenetic relationships on the basis of landmarks on the lower pharyngeal jaw bone. images in (B) are taken from dissected LPJs
(C) Results from the stomach and gut content analyses (in volume %).

(d) Explain the significance of the results found in Fig 8.2, 8.3 and 8.4 on the phylogeny of 'Neopro' and 'Cteben' species. [4]

[Total: 10]

Section B

Answer **one** question. Answer each part on a **separate** piece of paper.

Write your answers on separate answer paper provided.
Your answer should be illustrated by large, clearly labelled diagrams, where appropriate.
Your answer must be in continuous prose, where appropriate.
Your answer must be set out in sections **(a)**, **(b)** etc., as indicated in the question.

9 **(a)** Compare the lytic and lysogenic cycles of bacteriophages. [6]

(b) Describe the reproductive cycle of the influenza virus. [8]

(c) Distinguish prokaryotic and eukaryotic gene structure & regulation. [6]

[Total: 20]

10 **(a)** Describe how carbohydrates are formed in the Calvin cycle. [7]

(b) Outline the role of NAD and NADP in cells. [7]

(c) Compare and contrast the process of electron transport through the electron transport chain in the mitochondrion and chloroplast. [6]

[Total: 20]

END OF PAPER

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